James Foehner

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EDUCATION

Northeastern University, Khoury School of Computer Science *Boston, MA*
Candidate for BS in Data Science *Expected May 2027*
Coursework: Programming with Data Science, Machine Learning & Data Mining, Programming in C++, Statistics for Economists, Algorithms

SKILLS

Languages: Python | R | C++
Technical: Artificial Intelligence | Supervised & Reinforcement Learning | Predictive Modeling | LLM Fine-Tuning & Evaluation | SQL
Finance: Financial Modeling (Excel) | Valuation (Comps, Sum-of-the-Parts) | Event Studies | WRDS (CRSP, Compustat) | SEC EDGAR

PERSONAL PROJECTS

Activist Hedge Fund Target Screener & Post Holdings Activist Pitch *Oct 2026*
Python, SQL, WRDS (CRSP, Compustat), SEC EDGAR, Excel, Claude API | github.com/jamefoehner/activist-target-screener
- Built a Python pipeline collecting 232 activist campaigns (2010–26) from SEC 13D filings; an event study on CRSP returns found 7.0% abnormal returns around announcements (t = 13.3), with one-year gains concentrated in targets later acquired (+19% vs. ~0%)
- Trained a gradient-boosting model on 44K firm-years of peer-relative Compustat data with walk-forward validation; top 10% of predictions caught 27% of future targets (AUC 0.73), ranking E2open in the top 0.4% before Elliott's 2023 campaign
- Used the model to identify Post Holdings (ranked 7th of 2,600+) and built an Excel model comparing its debt-funded buybacks with a plan to sell Weetabix and delever (~$111/share vs. $77; stronger in a downside case)
- Tested two thesis assumptions against data (a deleveraging re-rating; spin-off value across 182 spin-offs since 2010), found neither held, and rebuilt the pitch around capital allocation in an investment memo
- Built an LLM evaluation benchmark on 100 hand-labeled 13Ds: Claude Sonnet with a labeling guide fully matched reference labels on 83% of filings (20 of 20 vs. 18 of 20 for a second human); model disagreement found a 7% label error rate

Reward Hacking in Reinforcement Learning for 13D Filing Analysis *Nov 2026*
Python, PyTorch, Hugging Face, LoRA, GRPO | github.com/jamefoehner/activist-target-screener
- Fine-tuned Qwen2.5-1.5B with LoRA on Claude-labeled 13D filings to tag ten activist demands; macro-F1 on 89 hand-labeled filings rose from 0.35 to 0.67
- Wrote a GRPO training loop and showed RL from the base model collapsed under both an accuracy reward (all-zero answers on 66 of 89 filings, 87% accuracy) and a set-overlap reward (single label, below a constant baseline)
- Used paired bootstrap tests to show that starting RL from the fine-tuned model avoided collapse, but its best gain (0.73 macro-F1) was not distinguishable from fine-tuning alone

NBA Prediction AI *Jan 2026 - Apr 2026*
- Built an end-to-end ML pipeline using 3 seasons of NBA play-by-play data (~80K rows, 30 features)
- Trained a Gradient Boosting Classifier for in-game win probability, achieving ROC-AUC 0.84 and Brier score 0.071, outperforming a Logistic Regression baseline by +0.08 AUC
- Developed a Random Forest play-type classifier across 6 imbalanced event classes, reaching 68% accuracy on a held-out test set using stratified splits and per-class evaluation
- Engineered rolling-window contextual features for three Random Forest regressors predicting player points, assists, and rebounds achieved ±4.1 pt MAE with 80% confidence intervals via individual tree variance.
- Deployed an interactive 3-tab Streamlit app with real-time Plotly visualizations for live model inference and sensitivity analysis

Airbnb Price Prediction Model *Aug 2025 - Dec 2025*
- Built an end-to-end ML pipeline on 3,500+ Boston Airbnb listings, covering data cleaning, outlier handling, and five engineered features, to predict nightly price.
- Benchmarked five regression models (Linear, Ridge, Lasso, Random Forest, Gradient Boosting) with GridSearchCV and cross-validation; the tuned Random Forest cut MAE from about $110 to $75 (about 32%) and raised test R^2 from 0.12 to 0.48 versus the linear baseline.
- Used feature importance and geospatial analysis to show that location and host listing count were the strongest price drivers and analyzed the train/test gap and residuals to diagnose overfitting and heteroscedasticity.

Aircraft Incident Database & Analytics *May 2025 - Jul 2025*
- Designed and implemented a 3NF relational database for aircraft incident data
- Developed an ERD and defined primary/foreign key relationships and database constraints.
- Built an ETL pipeline in R to load and transform CSV data into a cloud-hosted MySQL database using batched SQL inserts and transaction management.
- Developed SQL queries, views, and MySQL stored procedures to support incident reporting, airline analysis, and database business logic.
- Validated database integrity by comparing source CSV data against database records using counts, aggregates, and consistency tests

EXPERIENCE

Chart House – *Server* *Aug 2026 - Present*

- Serve guests in a high-volume, detail-oriented restaurant while managing multiple tables under time pressure
- Drive revenue by recommending premium menu items and wine pairings tailored to guest preferences